1. Equivalence.

(binary quad form) $f(x,y) = ax^2 + bxy + cy^2$

(discriminant) $d = b^2 - 4ac$

(principal form) $\begin{cases} x^2 - \frac{1}{4}dy^2 & d \equiv 0 \pmod 4 \\ x^2 + xy + \frac{1}{4}(1-d)y^2 & d \equiv 1 \pmod 4 \end{cases}$

$$f(x,y) = ax^2 + bxy + cy^2$$

then $4af(x,y) = (2ax+by)^2 - dy^2$

$d > 0$: f is positive (or negative) definite. i.e always same sign (or zero)

$d < 0$: f is indefinite. i.e takes both signs

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equivalence of two quad forms

$X = \begin{pmatrix} x \\ y \end{pmatrix}$

$X' = \begin{pmatrix} x' \\ y' \end{pmatrix}$

① quad form

$F = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}$

$f = X^T F X$

② integral unimodula substitution

$U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ $\begin{cases} \det U = 1 \\ p, q, r, s : \text{integers} \end{cases}$

(i) $X = UX'$

(ii) $F' = U^T F U$ (equivalent)

- reflexive (I), symmetric (U^{-1}), transitive (U_1U_2)
- values assumed by equivalent forms

as (x,y) runs through ~~integers~~ relatively prime integers are same.

Note.

$U^{-1} = \begin{pmatrix} s & -q \\ -r & p \end{pmatrix}$

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equivalent forms have same discriminant

$\because$ $\det F = ac - \frac{b^2}{4}$

$\det F' = \det U^T F U = 1 \cdot (ac - \frac{b^2}{4}) \cdot 1$

Properties of integral unimodula substitution

let $U = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ with $\begin{cases} \det U = 1 \\ p, q, r, s : \text{integers} \end{cases}$

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① U is bijection $(\mathbb{Z}^2 \longrightarrow \mathbb{Z}^2)$

(i) injection : $UX_1 = UX_2$ then $U^{-1}UX_1 = U^{-1}UX_2$

$X_1 = X_2$

(ii) surjection : $Y \in \mathbb{Z}^2$ then $X = U^{-1}Y \in \mathbb{Z}^2$

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② $X = UX'$ then

$$\begin{aligned} f'(X') &= X'^T F' X' \\ &= X'^T U^T F U X' \\ &= X^T F X = f(X) \end{aligned}$$

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③ $(x, y) = 1$ iff $(x', y') = 1$

$\because$ $x = px' + qy'$ then (x', y') divides both x, y

$y = rx' + sy'$ however $(x, y) = 1$

2. Reduction

positive definite quad form $\Rightarrow$ $\underbrace{d<0}_{\text{definite}}$ and $\underbrace{a>0}_{\text{positive}}$ $\Rightarrow$ $c>0$

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transform f into binary form s.t $|b| \le a \le c$

$F = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}$

① replace $a>c$ by $a<c$ $\quad U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ then $F' = \begin{pmatrix} c & -\frac{b}{2} \\ -\frac{b}{2} & a \end{pmatrix}$

② replace $|b|>a$ by $|b| \le a$ $\quad U = \begin{pmatrix} 1 & \pm 1 \\ 0 & 1 \end{pmatrix}$ then $F' = \begin{pmatrix} a & \frac{b \pm 2a}{2} \\ \frac{b \pm 2a}{2} & a \pm b + c \end{pmatrix}$

- process terminates $\because$ a decreasing!
- reduced forms

(i) $-a < b \le a < c$ $\quad$ $\because$ 최신 ② $\Rightarrow$ $F' = \begin{pmatrix} a & \frac{a}{2} \\ \frac{a}{2} & c \end{pmatrix}$

i.e $-a < b$

(ii) $0 \le b \le a = c$ $\quad$ $\because$ 최신 ① $\Rightarrow$ $F' = \begin{pmatrix} c & -\frac{b}{2} \\ -\frac{b}{2} & a \end{pmatrix}$

i.e if $a=c$ then $0 \le b$

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class number $h(d)$ $=$ # of reduced forms with discriminant d

$=$ # of inequivalent classes of binary quad forms with discriminant d

i.e any two reduced forms are not equivalent.

$h(d)$ is finite.

$\because$ $-d = 4ac - b^2 \ge 3ac$

then $|b|, a, c \le |d|/3$

e.g $d=-4$ $\Rightarrow$ $3ac \le 4$

$a=1,\ c=1$

$\Rightarrow$ $b=0$

$\Rightarrow$ $h(-4)=1$

Claim if $\underbrace{\text{reduced}}_{\substack{-a<b\le a\le c \\ 0\le b\le a=c}}$ form f, f' are $\underbrace{\text{equivalent}}_{F'=U^TFU}$ then $\begin{cases} a'=a \\ b'=\pm b \\ c'=c \end{cases}$

- if $|x|\ge|y|$ (x, y : non-zero integers)

then $f(x,y) \ge |x|(a|x|-|by|)+c|y|^2$

$\ge |x|^2(a-|b|)+c|y|^2$

$\ge a-|b|+c$

- smallest values assumed by f, f' with $(x,y)=1$

$(\pm1, 0) \xrightarrow{f,f'} a$

$(0, \pm1) \longrightarrow c$

$(\pm1,\pm1)$ or $(\pm1,\mp1) \longrightarrow a-|b|+c$

$\Rightarrow a'=a,\ b'=\pm b,\ c'=c$

Claim if $b'=-b$ then $b=0$

- consider only $-a<b<a<c$

because (i) $-a<b$ (reduced f)

$-a<-b$ (reduced f')

(ii) $a=c$ then $b\ge 0$ (reduced f)

$-b\ge 0$ (reduced f')

- $f(x,y)\ge a-|b|+c>c>a$ (x, y : non-zero integers)

- $U=\begin{pmatrix} p & q \\ r & s \end{pmatrix}$ then

$a=f'(\pm1,0)=f(p,r) \rightsquigarrow p=\pm1,\ r=0$

$c=f'(0,\pm1)=f(q,s) \rightsquigarrow q=0,\ \underbrace{s=\pm1}_{ps-qr=1}$

- $U=\begin{pmatrix} \pm1 & 0 \\ 0 & \pm1 \end{pmatrix}$ then $F'=U^TFU=F$

3. Representations by binary forms

"properly represented"

$n = \underbrace{f(x,y)}_{\text{binary form}}$ with $(x,y)=1$ $(\exists f)$ iff $x^2 \equiv d \pmod{4n}$ is solvable

$\Gamma\!\cdot\!\cdot$

- $b^2 \equiv d \pmod{4n}$
- $d = b^2 - 4\cancel{a}\,n\,c$
- $f(1,0) = n$

$\Gamma\!\cdot\!\cdot$

- $n = f(p,r)$ with $(p,r)=1$
- $U = \begin{bmatrix} p & s \\ r & s \end{bmatrix}$ with $F' = U^T F U$

  $f'(1,0) = n$

  $\det U = 1$
- $\det F = \det F'$

  therefore $b'^2 - 4nc = d$

$(n,d)=1$ then ωs = # of proper representations of n by all reduced forms with discriminant d

- s : # of solutions of $x^2 \equiv d \pmod{4n}$ $(0 \le x < 2n)$
- ω : # of $\underbrace{\text{automorphs}}_{U \text{ that takes } F \text{ into itself}}$ of reduced form

$$\omega = \begin{cases} 2 & (d < -4) \quad \text{i.e } U = \pm I \\ 4 & (d = -4) \\ 6 & (d = -3) \end{cases}$$

related to Pell equation.

4. Sums of two squares

$n = x^2+y^2$ iff every prime p with $\begin{cases} p \mid n \\ p \equiv 3 \pmod 4 \end{cases}$

$(\exists x, y \in \mathbb{Z})$ occures to even power in standard factorization.

- $p \mid x^2+y^2$ with $p \equiv 3 \pmod 4$
- $x^2 \equiv -y^2 \pmod p$
- $\left(\frac{-1}{p}\right) = -1$ (quadratic non-residue) therefore $p \nmid x$, $p \nmid y$
- $(x/p)^2 + (y/p)^2 = n/p^2 \Rightarrow$ induction.

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claim if (i) n is square free

(ii) each odd prime divisor p of n satisfies $p \equiv 1 \pmod 4$ $\Rightarrow n = x^2+y^2$.

i.e $n = p_1^{\equiv 1} p_2^{\equiv 1} \cdots p_k^{\equiv 1} \pmod 4$

- if this holds, $nm^2 = (xm)^2 + (ym)^2$
- x^2+y^2 is the only reduced form with discriminant -4
- n is properly represented by x^2+y^2 iff $x^2 \equiv -4 \pmod{4n}$
- $\left(\frac{-1}{p}\right) = +1$ (quad-residue) $\xrightarrow{\text{CRT}}$ $y^2 \equiv -1 \pmod n$ i.e quad residue for each prime divisor $\Longrightarrow (2y)^2 \equiv -4 \pmod{4n}$

$(x^2+y^2)(x'^2+y'^2) = (xx'+yy')^2 + (xy'-yx')^2$

$\Rightarrow$ consider only prime values of n

- # of representation of n as x^2+y^2 $= 4 \sum \left(\frac{-1}{m}\right)$

$\Rightarrow$ prime $p \equiv 1 \pmod 4$ can be expressed in 8 ways as x^2+y^2

5. Sums of four Squares

Note.

$2 = 1^2 + 1^2 + 0^2 + 0^2$

- every natural number is the sum of 4 integer squares

- $(x^2+y^2+z^2+w^2)(x'^2+y'^2+z'^2+w'^2)$

$$= (xx'+yy'+zz'+ww')^2 + (xy'-yx'+wz'-zw')^2$$
$$+ (xz'-zx'+yw'-wy')^2 + (xw'-wx'+zy'-yz')^2$$

- consider odd prime p.

- mutually incongruent $\begin{cases} x^2 & (0 \le x \le \frac{1}{2}(p-1)) \\ -1-y^2 & (0 \le y \le \frac{1}{2}(p-1)) \end{cases}$

- $x^2 \equiv -1-y^2 \pmod p$

with $x^2+y^2+1 < 1 + 2(\frac{1}{2}p)^2 < p^2$

- $mp = x^2+y^2+1$ with $0 < m < p$

- l is least positive integer with $lp = x^2+y^2+z^2+w^2$ $\Rightarrow l \le m < p$

- l is odd $\because$ otherwise (i) even number of x, y, z, w is odd

(ii) $\frac{1}{2}lp = (\frac{1}{2}(x+y))^2 + (\frac{1}{2}(x-y))^2 + (\frac{1}{2}(z+w))^2 + (\frac{1}{2}(z-w))^2$

Claim $l = 1$ otherwise

(i) x', y', z', w' be numerically least residue $\pmod l$

(ii) $n = x'^2 + y'^2 + z'^2 + w'^2$

(iii) $n \equiv 0 \pmod l$ with (i) $0 < n$ $\because$ otherwise $l \mid p$

(ii) $n < 4(\frac{1}{2}l)^2 = l^2$ $\because$ l is odd

i.e $n = kl$ with $0 < k < l$

(iv) $(kl)(lp)$ is the sum of 4 integer square.

with each squares divisible by l^2 $\Rightarrow$ kp sum of 4 squares

$\rightarrow\leftarrow$ because $0 < k < l$

Legendre — Gauss

$n = x^2 + y^2 + z^2 \iff n \neq 4^j(8k+7)$

$(n \in \mathbb{N})$

$\because (\Rightarrow)$ $x^2 \equiv 0, 1, 4 \pmod 8$

Waring (1770)

$g(k)$

natural number = sum of 4 squares

= sum of 9 cubes

= sum of 19 biquadrates

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Claim for every integer $k \geq 2$

(i) integer s s.t. every natural number

$= x_1^k + \cdots + x_s^k$

(ii) $g(k)$ is the least such s.

$\because$ Hilbert (1909)

$\because$ Hardy – Littlewood (1920)

- $f(z) = 1 + z^{1^k} + z^{2^k} + \cdots = \sum_{m=0}^{\infty} z^{m^k}$

- $(f(z))^s = \left(\sum_{m_1=0}^{\infty} z^{m_1^k}\right)\left(\sum_{m_2=0}^{\infty} z^{m_2^k}\right) \cdots \left(\sum_{m_s=0}^{\infty} z^{m_s^k}\right)$

$= \sum_{n=0}^{\infty} r(n) \cdot z^n$

$\because$ $n = m_1^k + m_2^k + \cdots + m_s^k$

- $r(n)$: # of representation of n

in required form

- $r(n) = \dfrac{1}{2\pi i} \displaystyle\int_C \frac{(f(z))^s}{z^{n+1}}\, dz$ $\Rightarrow$ delicate division of contour $\begin{cases} \text{major arcs} \\ \text{minor arcs} \end{cases}$